

Applications Development and Emerging Technologies



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MODULE 3

PHP Arrays and User Defined Functions



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Objectives

- *To know the different approach of using arrays in PHP.*
- *To be familiar with using print_r function and var_dump function for array inspection.*
- *To use PHP lazy function foreach to iterate through array elements.*
- *To implement one dimensional and multi-dimensional in the program.*
- *To know what is associative array in PHP.*
- *To use different predefined functions in manipulating array elements.*
- *To defined what is function and to use function keyword in declaring function in PHP.*
- *To create a user defined function using different approach.*



Objectives (continue)

- *To know use variables that is globally declared and locally declared.*
- *To create static variables that can be used in accumulating data every time the function calls.*

PHP Arrays



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PHP Arrays and User Defined Functions

Array

is used to aggregate a series of similar items together, arranging and dereferencing them in some specific way.

- Each member of the array index references a corresponding value and can be a simple numerical reference to the value's position in the series, or it could have some direct correlation to the value.
- PHP array does not need to declare how many elements that the array variable have.
- Array index in PHP can be also called as array keys.
- Array can be used as ordinary array same as in C and C++ arrays.



PHP Arrays and User Defined Functions

Array: one dimensional

Example: Code 1

```
<?php
    $fruit[0] = "orange";
    $fruit[1] = "apple";
    $fruit[2] = "grapes";
    $fruit[3] = "banana";
    echo "<pre>";
    print_r($fruit);
    echo "</pre>";
?>
```

Example: Code 2

```
<?php
    $fruit[] = "orange";
    $fruit[] = "apple";
    $fruit[] = "grapes";
    $fruit[] = "banana";
    echo "<pre>";
    print_r($fruit);
    echo "</pre>";
?>
```

Example: Code 3

```
<?php
    $fruit = array("orange", "apple", "grapes", "banana");
    echo "<pre>";
    print_r($fruit);
    echo "</pre>";
?>
```

Output:

```
Array
(
    [0] => orange
    [1] => apple
    [2] => grapes
    [3] => banana
)
```



PHP Arrays and User Defined Functions

Array: Functions for visualizing arrays

`print_r()` function

short for print recursive. This takes an argument of any type and prints it out, which includes printing all its parts recursively.

`var_dump()` function

is same as the `print_r` function except that it prints additional information about the size and type of the values it discovers

`print_r()` and `var_dump()` functions are commonly used for debugging. The point of this of these functions is to help you visualize what's going on with compound data structures like arrays.



PHP Arrays and User Defined Functions

Array: Functions for visualizing arrays

Example: using `print_r()`

```
<?php
    $fruit = array(
        "orange",
        "apple",
        "grapes",
        "banana");
    echo "<pre>";
    print_r($fruit);
    echo "</pre>";
?>
```

Output:

```
Array
(
    [0] => orange
    [1] => apple
    [2] => grapes
    [3] => banana
)
```

Example: using `var_dump()`

```
<?php
    $fruit = array(
        "orange",
        "apple",
        "grapes",
        "banana");
    echo "<pre>";
    var_dump($fruit);
    echo "</pre>";
?>
```

Output:

```
array(4) {
    [0]=>
    string(6) "orange"
    [1]=>
    string(5) "apple"
    [2]=>
    string(6) "grapes"
    [3]=>
    string(6) "banana"
}
```



PHP Arrays and User Defined Functions

Array: Looping through array elements

`foreach()` function

is a statement used to iterate or loop through the element in an array. With each loop, a `foreach` statement moves to the next element in an array.

`foreach` statement

specify an array expression within a set of parenthesis following the `foreach` keyword.

Syntax

```
foreach($arr as $value){  
    //do something with $value variable  
}  
  
foreach($arr as $key => $value){  
    //do something with $key and/or $value variable  
}
```



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PHP Arrays and User Defined Functions

Array: Looping through array elements

<code>\$arr</code>	The name of the array that you're walking through.
<code>\$key</code>	The name of the variable where you want to store the key. (optional)
<code>\$value</code>	The name of the variable where you want to store the value

Example:

```
<?php
    $fruit = array(
        "orange",
        "apple",
        "grapes",
        "banana",
        "dragon fruit");
    foreach($fruit as $value){
        echo "$value<br/>";
    }
?>
```

Output:

```
orange
apple
grapes
banana
dragon fruit
```



PHP Arrays and User Defined Functions

Array: Looping through array elements

Example:

```
<?php
    $person = array(
        'First Name' => 'Juan',
        'Last Name' => 'Dela Cruz',
        'Date of Birth' => 'April 3, 1986',
        'Gender' => 'Male'
    );

    foreach($person as $value){
        echo "$value <br/>";
    }
    echo "<p/>";
    foreach($person as $key => $value){
        echo "$key: $value <br/>";
    }
?>
```

Output:

Juan
Dela Cruz
April 3, 1986
Male

First Name: Juan
Last Name: Dela Cruz
Date of Birth: April 3, 1986
Gender: Male



PHP Arrays and User Defined Functions

Arrays: Multidimensional arrays

Example:

```
<?php
$person = array(
    'Leader' => array(
        'First Name' => 'Juan',
        'Last Name' => 'Dela Cruz',
        'Date of Birth' => 'April 3, 1986',
        'Gender' => 'Male' ),
    'Assistant' => array(
        'First Name' => 'Rica',
        'Last Name' => 'San Pedro',
        'Date of Birth' => 'January 8, 1980',
        'Gender' => 'Female' ),
    'Member' => array(
        'First Name' => 'Jose',
        'Last Name' => 'Dela Paz',
        'Date of Birth' => 'March 2 1985',
        'Gender' => 'Male' )
);

foreach($person as $pkey => $parr){
    echo "<h2>$pkey</h2>";
    foreach($parr as $key => $value){
        echo "$key: $value<br/>";
    }
}
```

Output:

Leader

First Name: Juan
Last Name: Dela Cruz
Date of Birth: April 3, 1986
Gender: Male

Assistant

First Name: Rica
Last Name: San Pedro
Date of Birth: January 8, 1980
Gender: Female

Member

First Name: Jose
Last Name: Dela Paz
Date of Birth: March 2 1985
Gender: Male

PHP Arrays and User Defined Functions

Arrays: Sorting

Function	Description
<code>sort(\$array)</code>	Sorts by value; assign new numbers as the keys
<code>rsort(\$array)</code>	Sorts by value in reverse order; assign new number as the keys
<code>asort(\$array)</code>	Sorts by value; keeps the same key
<code>arsort(\$array)</code>	Sorts by value in reverse order; keeps the same key
<code>ksort(\$array)</code>	Sorts by key
<code>krsort(\$array)</code>	Sorts by key in reverse order
<code>usort(\$array, functionname)</code>	Sorts by a function



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PHP Arrays and User Defined Functions

Arrays: Sorting

Example:

```
<?php
    $fruit = array(
        "orange", "apple",
        "grapes", "banana",
        "guava", "blue berry");

    sort($fruit); //sort array in ascending
    echo "<pre>";
    print_r($fruit);
    echo "</pre>";

    rsort($fruit); //sort array in descending
    echo "<pre>";
    print_r($fruit);
    echo "</pre>";
?>
```

Output:

```
Array
(
    [0] => apple
    [1] => banana
    [2] => blue berry
    [3] => grapes
    [4] => guava
    [5] => orange
)

Array
(
    [0] => orange
    [1] => guava
    [2] => grapes
    [3] => blue berry
    [4] => banana
    [5] => apple
)
```



PHP Arrays and User Defined Functions

Arrays: Sorting

Example:

```
<?php
    $fruit = array(
        "orange", "apple",
        "grapes", "banana",
        "guava", "blue berry");

    asort($fruit);
    echo "<pre>";
    print_r($fruit);
    echo "</pre>";

    arsort($fruit);
    echo "<pre>";
    print_r($fruit);
    echo "</pre>";
?>
```

Output:

```
Array
(
    [1] => apple
    [3] => banana
    [5] => blue berry
    [2] => grapes
    [4] => guava
    [0] => orange
)

Array
(
    [0] => orange
    [4] => guava
    [2] => grapes
    [5] => blue berry
    [3] => banana
    [1] => apple
)
```



PHP Arrays and User Defined Functions

Arrays: Sorting

Example:

```
<?php
$fruit = array(
    '1010' => "orange",
    '1002' => "apple",
    '1023' => "grapes",
    '1006' => "banana",
    '1015' => "guava",
    '1009' => "blue berry");

ksort($fruit);
echo "<pre>";
print_r($fruit);
echo "</pre>";

krsort($fruit);
echo "<pre>";
print_r($fruit);
echo "</pre>";
```

?>

Output:

```
Array
(
    [1002] => apple
    [1006] => banana
    [1009] => blue berry
    [1010] => orange
    [1015] => guava
    [1023] => grapes
)

Array
(
    [1023] => grapes
    [1015] => guava
    [1010] => orange
    [1009] => blue berry
    [1006] => banana
    [1002] => apple
)
```



PHP User Defined Functions



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PHP Arrays and User Defined Functions

Functions

is a group of PHP statements that performs a specific task. Functions are designed to allow you to reuse the same code in different locations.

User defined functions

functions that are provided by the user of the program.

Predefined functions

functions that are built-in into PHP to perform some standard operations



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PHP Arrays and User Defined Functions

Functions: User defined functions

Syntax

```
function name(param) {  
    //code to be executed by the function  
}
```

Where

function - is the keyword used to declare a function

name - is the name of the function or function identifier

param - is the formal parameters of the function.

Parameter must follow the rule of naming identifier.



PHP Arrays and User Defined Functions

Functions: Function with no parameters

Example:

```
<?php
function redline() {
    ?>
    <hr style="
background-color:#f00;
height:5px;
    " />
<?php
}

function footer() {
    echo "PHP Test Page &copy 2013";
}

echo "First Text";
redline();
echo "Second Text";
redline();
footer();
?>
```

Output:

First Text

Second Text

PHP Test Page © 2013



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PHP Arrays and User Defined Functions

Functions: Function with parameters

Example:

```
<?php
function hrline($height, $color){
    ?>
    <hr
        style="
            height:<?php echo $height; ?>;
            background-color:<?php echo $color ?>"
        />
    <?php
}

echo "First Text";
hrline("5px", "#00f");
echo "second Text";
hrline("10px", "#0f0");
?>
```

Output:

First Text



second Text



PHP Arrays and User Defined Functions

Functions: Function that returns a value

Example:

```
<?php
function qoutient($x, $y){
    return $x/$y;
}

function remark($grade){
    if($grade>=75){
        return true;
    } else {
        return false;
    }
}

printf("The qoutient is %.2f<br/>",qoutient(7,3));
if(remark(80)){
    echo "passed";
} else {
    echo "failed";
}

?>
```

Output:

The qoutient is 2.33
passed



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PHP Arrays and User Defined Functions

Functions: Nested function

Example:

```
<?php
function displayPerson() {
    function name() {
        echo "Juan Dela Cruz";
    }
}
```

Output:

```
name();
```

Fatal error: Call to undefined function name() in C:\wamp\www\php\sample11.php on line 8

```
?>
```

Example:

```
<?php
function displayPerson() {
    function name() {
        echo "Juan Dela Cruz";
    }
}
displayPerson();
name();
```

```
?>
```

Output:

Juan Dela Cruz



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PHP Arrays and User Defined Functions

Functions: Variable scope

Global Variables

is one that declared outside a function and is available to all parts of the program.

Local Variables

is declared inside a function and is only available within the function in which it is declared.

Static Variables

is used to retain the values calls to the same function.



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PHP Arrays and User Defined Functions

Functions: using variables

Example:

```
<?php
    $str = "This is global variable";
    function f(){
        echo "function f() was called<br>";
        echo "$str<br>";
    }
    f();
?>
```

Output:

function f() was called

Example:

```
<?php
    $str = "This is global variable";
    function f(){
        global $str;
        echo "function f() was called<br>";
        echo "$str<br>";
    }
    f();
?>
```

Output:

function f() was called
This is global variable



PHP Arrays and User Defined Functions

Functions: using variables

Example:

```
<?php
function f(){
    $str = "This is local variable";
}
f();
echo "Displaying local variable: ";
echo $str;
?>
```

Output:

Displaying local variable:

Example:

```
<?php
function f(){
    global $str;
    $str = "This is local variable";
}
f();
echo "Displaying local variable: ";
echo $str;
?>
```

Output:

Displaying local variable: This is local variable



PHP Arrays and User Defined Functions

Functions: using variables

Example:

```
<?php
function counter($value) {
    static $count = 0;
    $count+=$value;
    echo "The value of the counter is $count<br>";
}

counter(4);
counter(3);
counter(8);
counter(6);
?>
```

Output:

```
The value of the counter is 4
The value of the counter is 7
The value of the counter is 15
The value of the counter is 21
```



Summary

- *Array is used to aggregate a series of similar items together.*
- *Array index references a corresponding value.*
- *Array index can be simple numerical or have some direct correlation to the value.*
- *Array index is also known as Array Keys.*
- *print_r function is used to print the array structure.*
- *var_dump function is same as print_r function except it adds additional information about the data of each element.*
- *The foreach statement is use to iterate through the element in an array.*
- *Using foreach statement you can display both the keys and value of each element in the array.*
- *PHP provides functions for array manipulation such as sort(), rsort(), asort(), arsort(), ksort(), krsort(), and usort() functions.*

Summary (continue)

- *sort(), asort(), and ksort() functions are used to sort elements in the array in ascending order.*
- *rsort(), arsort(), and krsort() functions are used to sort elements in the array in descending order.*
- *sort() and rsort() does not maintain its index reference for each values.*
- *asort(), ksort(), arsort(), and krsort() maintains its reference for each values.*
- *asort() and arsort() used to sort elements by values.*
- *ksort() and krsort() used to sort elements by keys.*
- *Functions is a group of PHP statements that performs a specific task.*
- *Functions can be user defined generally defined by the user of the program and predefined that are build in using libraries.*

Summary (continue)

- *You use functions in different ways. Function can only do something without passing values. You can pass values to a function and you can ask functions to return a value.*
- *function keyword is used in PHP to declare a function.*
- *A function that is declared inside a function is said to be hidden.*
- *To gain access to a variable that is outside from the function we use the global keyword.*
- *We use static keyword to declare a variable inside a function that will act as accumulator variable this will let the program remember the last value of the variable that was used.*



Q&A SESSION

